LLL-L-51C
November 3, 1972
SUPERSEDING
Fed. Spec. LLL-L-51B
January 9, 1963

FEDERAL SPECIFICATION

LADDER, PORTABLE, WOOD: EXTENSION AND STRAIGHT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wood, single and extension, portable, rung ladders.

1.2 Classification.

1.2.1 Types and sizes. The ladders shall be of the following types and sizes, as specified (see 6.2):

Type I - Single ladder.
Length 8 through 30 feet by intervals of 2 feet.
Type II - Two section extension ladder.
Length 20 through 60 feet by intervals of 4 feet.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

T-R-650 - Rope, Yarn and Twine, Base Fiber.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coated, Electrodeposited, Requirements for.
TT-W-572 - Wood Preservative: Water-Repellent.
PPP-C-650 - Crates, Wood, Open and Covered.

FSC 5440

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Federal Standards

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cummulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1363 - Measurement of Wood Moisture Content.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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American National Standards Institute, Inc., (ANSI) Standard:

A14.1 - Safety Code for Portable Wood Ladders.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies hould be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The ladders shall conform to ANSI A14.1 and as specified herein.

3.2 First article (first-produced ladder). The supplier shall furnish one or more ladders for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce ladders that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Lumber species. Lumber species shall conform to group 1, 2, or 3 of ANSI A14.1 as specified herein.

3.3.2 Hardware.

3.3.2.1 Locks and guides. Locks and guides shall be made from steel or malleable iron.

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3.3.2.2 Pulleys. Pulleys shall be made from aluminum, cast iron, malleable iron or steel.

3.4 Cadmium plating and galvanizing. Ferrous metal hardware shall be cadmium plated in accordance with QQ-P-416, class 2, type III; or galvanized in accordance with QQ-Z-325, type III, class 2.

3.5 Moisture content of lumber. When tested as specified in 4.5.2.2, the moisture content of the lumber shall be not more than 15 percent.

3.6 Side rails.

3.6.1 Warp. Warp (bow or crook) in any side rail shall be not greater than 1/16 inch for each 2 feet of length of side rail.

3.6.2 Reinforcement. The side rails of each type I ladder and the extension section of type II ladders shall be reinforced with a 3/16-inch-diameter rivet located approximately 2-3/8 inches from each end of each side rail. The bottom section of each type II ladder shall be reinforced with 3/16-inch-diameter rivets located approximately 2-3/8 inches and 16-5/8 inches from each end of each side rail. The driven head of the rivets shall be driven over 1/2 inch OD washers and both rivet heads shall be not less than 7/16 inch in diameter (see figure 1).

3.6.3 Tie rods. Unless otherwise specified (see 6.2), side rails shall be held together with threaded tie rods with adjusting nuts with washers and lockwashers. When the ends of the tie rods are peened to retain the nuts, lockwashers are not required. When specified, the rungs shall be nailed to the side rails in accordance with ANSI A14.1. Tie rods shall be not less than 3/16-inch in diameter. The tie rods shall be placed directly under the end rungs in each straight ladder, and under the bottom and top rung of each section of extension ladders, and under rungs 4,7, 10, etc., in each section of each extension ladder.

3.6.4 Construction. Ladder side rails shall be constructed of group 2 or 3 woods.

3.7 Rungs. Rungs shall be made from lumber species group 1 in accordance with ANSI A14.1 and shall be not less than 1-1/8 inches in diameter for lengths up to 36 inches between side rails, and 1-1/4 inches in diameter for lengths from 36 through 72 inches. Rungs 28 inches and longer shall be provided with a center bearing. Tenons shall be not less than 7/8 inch in diameter.

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3.8 Nonslip bases. When specified (see 6.2), the ladders shall be equipped with nonslip bases.

3.9 Leveling devices. When specified (see 6.2), the ladders shall be equipped with leveling devices.

3.10 Hardware.

3.10.1 Guide irons. The minimum dimensions of guide irons shall be as shown on table I.

TABLE I. Dimensions or guides, Type II ladders

Size of ladder	Top guide Minimum		Bottom guide Minimum	
Feet	Inches		Inches	
	Thickness	Width	Thickness	Width
Through 40	3/16	by 1-1/2	3/16	by 1-1/4
Over 40	3/16	by 2	3/16	by 1-1/4

3.10.2 Locks. Spring locks shall be self-releasing and capable of sustaining a load of 1,000 pounds without failure when tested as specified in 4.5.2.1. Locks shall operate freely and positively.

3.10.3 Rope and pulley. The extension ladders shall be equipped with a rope and pulley for raising and lowering the upper section of the ladder. The pulley shall be not less than 1-1/4 inches inside diameter, and shall be attached so as not to weaken any part of the ladder. The rope shall conform to T-R-650, type I, class 2, and be of 1-inch circumference.

3.11 Preservative treatment. When specified (see 6.2), the wood components of each type and size ladder shall be preservative treated prior to assembly in accordance with TT-W-572, composition D.

3.12 Identification marking. The ladders shall be identified in accordance with MIL-STD-130.

3.13 Workmanship. The rungs and side rails of the ladders shall be smooth and free from splinters, and shoulders of rungs shall bear on the side rails. Guide irons shall be securely attached to the upper section in such a manner that the upper section cannot tip or fall out of the ladder while being raised or lowered or while in use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First-produced ladder inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First-produced ladder inspection.

4.3.1 Examination. The first-produced ladders shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced ladders shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all ladders of the same type and size submitted for inspection at one time.

4.4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.3 Examination.

4.4.3.1 Individual. Each type and size of ladder shall be examined for the critical defects specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

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4.4.3.2 Samples. Samples selected in accordance with 4.4.2 shall be examined for the defects specified in 4.5.1. AQL shall be 4.0 percent defective.

4.4.4 Test samples.

4.4.4.1 Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The ladders shall be examined as specified herein for the following defects:

Critical

1. Lumber characteristics of ANS A14.1 not as specified.
2. Cracks or splits in ladder rungs.
3. Rungs not as specified.

Major

101. Material not as specified.
102. Cadmium plating or galvanizing of ferrous metal components not as specified.
103. Moisture content of lumber not as specified.
104. Warp of side rails greater than that specified.
105. Reinforcement of side rails not as specified.
106. Tie rods not as specified (when applicable).
107. Construction not as specified.
108. Nonslip base not as specified (when applicable).
109. Leveling devices not as specified (when applicable).
110. Guide irons not as specified on extension ladders.
111. Ladder locks for extension ladders not as specified.
112. Ropes and pullys not as specified.
113. Finish and preservative treatment not as specified.
114. Identification marking missing, incomplete or not legible.
115. Workmanship not as specified.

4.5.2 Tests methods.

4.5.2.1 Ladder locks. Attach the lock to one end of a wooden beam of suitable strength using fasteners used by the supplier to attach locks to ladders. The hook of the lock shall extend just far enough beyond the beam

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to permit insertion of a 1-inch bolt or pin to support the test load. Support the beam by a 1-inch bolt through the opposite end from the lock (see figure 2). Suspend a test load of 1,000 pounds for a period of 5 minutes. Any evidence of permanent deformation, rupture, elongation, or cracking shall constitute failure of this test.

4.5.2.2 Moisture content of lumber. The lumber shall be tested for moisture content in accordance with MIL-STD-1363, method optional. Nonconformance to 3.5 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

116.	Extension ladders not retracted and secured as specified for level A.
117.	Rope not removed, packaged, and secured as specified for level A.
118.	Crate not as specified for level A or B.
119.	Ladders packed together exceed crate limitations for level A or B.
120.	Shroud not provided for level A.
121.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Extension ladders shall be retracted to minimum length and secured to prevent movement. The rope shall be removed and packaged in a waterproof bag conforming to MIL-B-117, type I, class B, style 1. The packaged rope shall be secured to the ladder by stapling beyond the heat seal in a manner to prevent destroying the waterproof nature of the bag.

5.1.2 Level C. The ladders shall be packaged in a manner to assure protection against deterioration and damage from the supplier to the initial destination.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The ladders shall be packed in a close-fitting crate conforming to PPP-C-650, type II, style A. Ladders shall be packed individually or ladders of like description shall be packed together in quantities not to exceed the weight or dimension limitations of the crate. The weight or dimensional limitations of the crate shall be waived when the weight or dimensions of an individual ladder exceeds that limitation. The ladder shall be blocked, braced, anchored, and protected with a shroud in accordance with the appendix to the crate specification.

5.2.2 Level B. The ladders shall be packed as specified in 5.2.1 for level A except a shroud shall not be required.

5.2.3 Level C. The ladders shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military agencies. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with FED. STD. No. 123.

6. NOTES

6.1 Intended use. Wood ladders are intended to be used where electrical safety hazard may exist and metal ladders cannot be used.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Type and length of ladder required (see 1.2).
(c) Time frame required for submission of first-produced ladder, and number of ladders required (see 3.2).
(d) When tie rods are not required (see 3.6.3).
(e) When nailing of rungs is required (see 3.6.3).
(f) When nonslip bases are required (see 3.8).

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(g) When leveling devices are required (see 3.9).
(h) When the wood components shall be preservation treated (see 3.11)
(i) Level of packaging and level of packing required (see 5.1 and 5.2).

6.3 First-produced ladder. Any changes or deviations of production ladders from the approved first-produced ladder during production will be subject to approval of the contracting officer. Approval of the first-produced ladder will not relieve the supplier of his obligation to furnish ladders conforming to this specification.

MILITARY INTEREST

Custodians:

Army - ME
Air Force - 84

User activities:

Navy - OS, MC, SH, YD

Preparing activity:

Army - ME

CIVIL AGENCIES INTEREST

GSA - FSS
AFS

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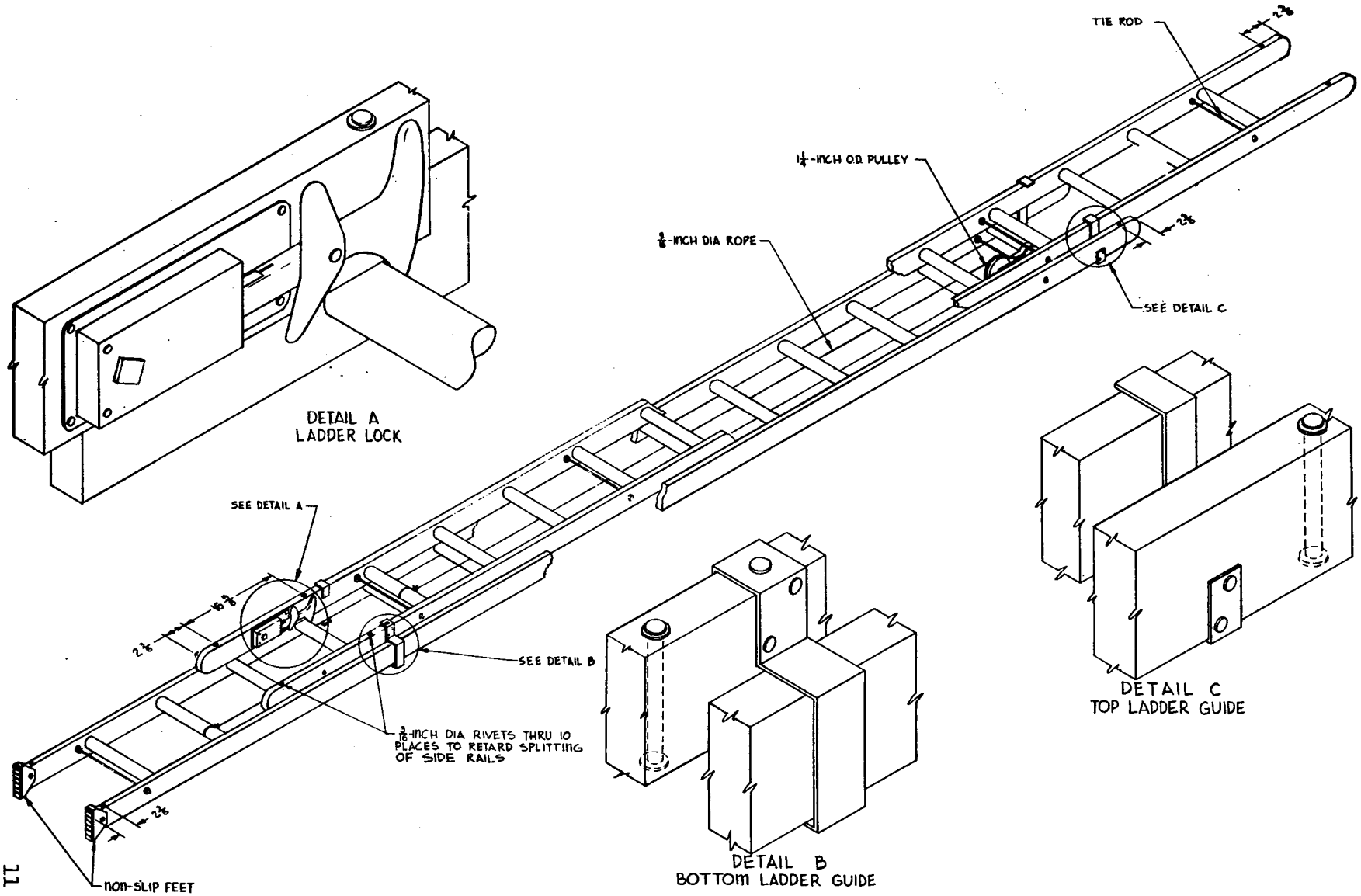


FIGURE 1. LADDER DETAILS

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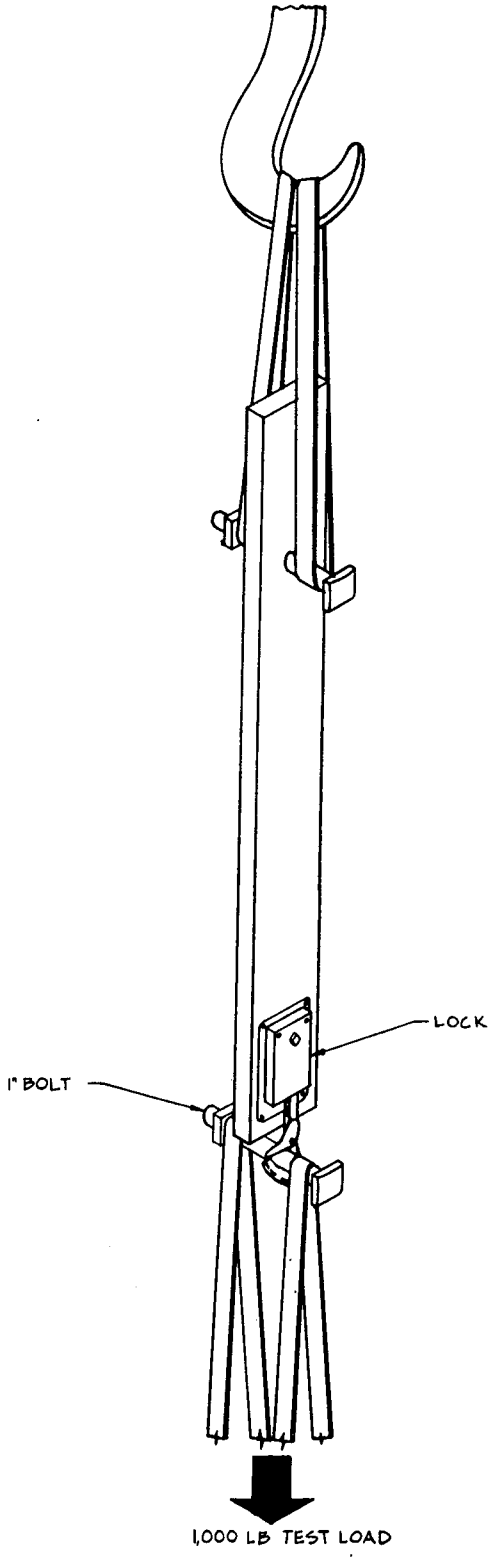


FIGURE 2. LOCK TEST SCHEME

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION LLL-L-51C - Ladder, Portable, Wood: Extension And Straight

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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